



IIT DELHI  
Indian Institute Of  
Technology Delhi

# GRAPH NEURAL NETWORKS FOR AUTISM SPECTRUM DISORDER

Modeling brain connectivity as graphs to  
study and classify Autism Spectrum  
Disorder using Graph Neural Networks



# INTERNSHIP PRESENTATION



Indian Institute of Technology, Delhi

Under the Guidance of

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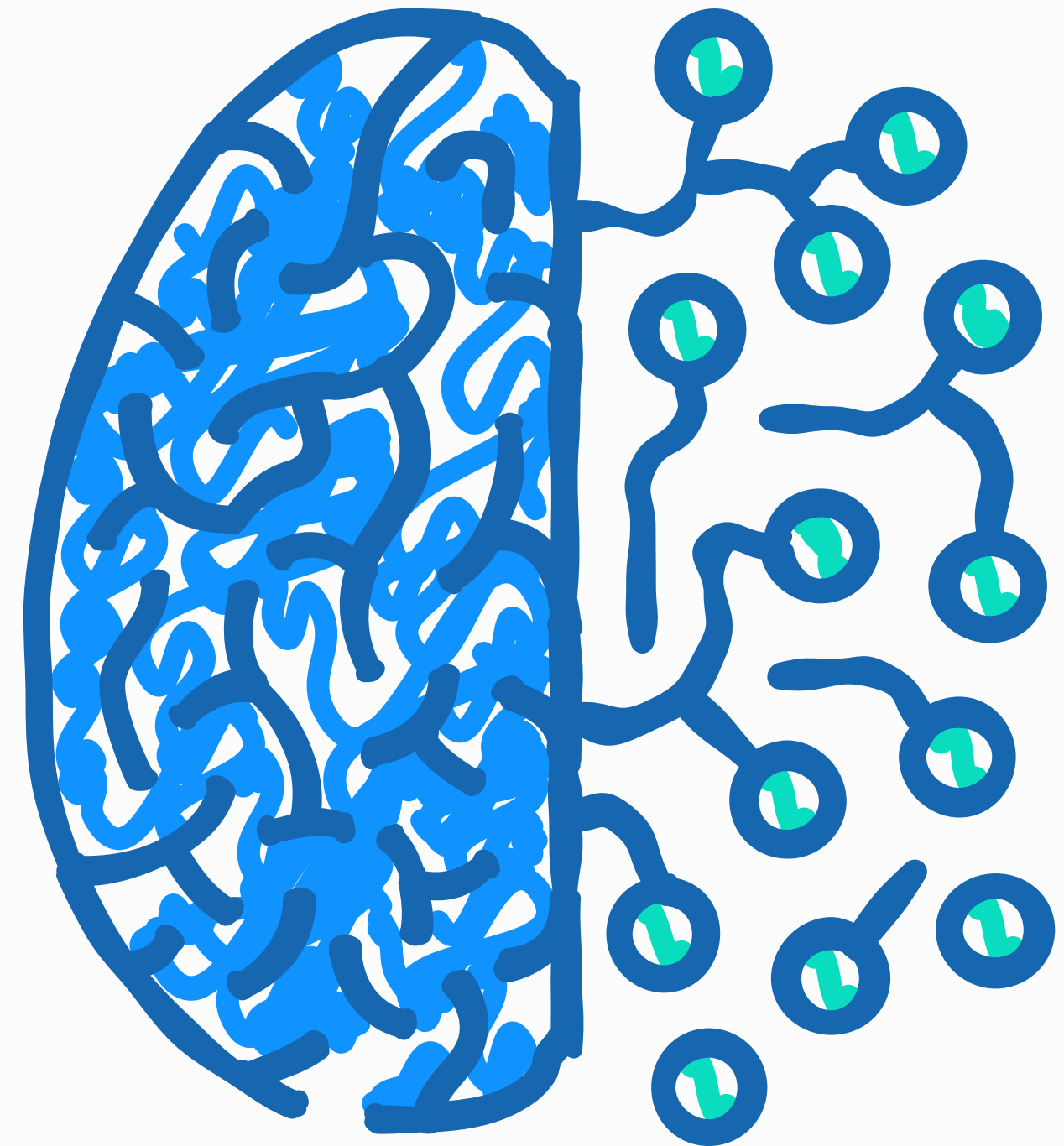
Presented By

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Summer Intern  
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# ABSTRACT

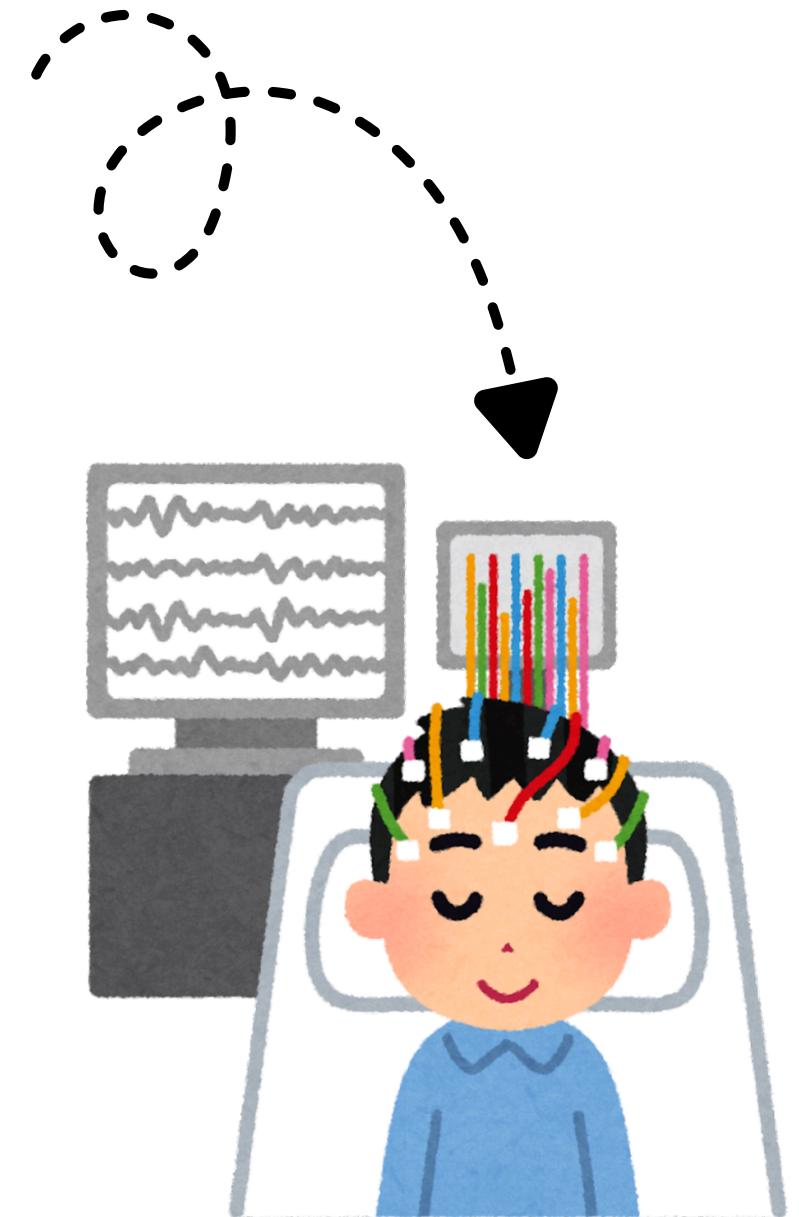
This internship focused on applying Graph Neural Networks (GNNs) to resting-state fMRI data for Autism Spectrum Disorder (ASD) classification using the ABIDE dataset. The work began with a from-scratch implementation of a 2-layer GCN on the Cora citation network to build a foundational understanding of graph convolutions and message passing. This was followed by a literature review of six papers spanning GNN benchmarking, learnable brain-network generation (FBNetGen), demographic-aware GNNs (BrainADNet), spatiotemporal GNNs (GLNSTGNN), attention-based brain transformers, and signed graph convolutions for handling anticorrelated connectivity. Building on this, FBNetGen was implemented end-to-end on the ABIDE dataset to generate task-aware brain graphs instead of relying on fixed Pearson-correlation adjacency matrices. Finally, a dedicated study was conducted on the latent space learned by a GCN classifier - examining, via PCA and UMAP, whether ASD/HC groups, acquisition sites, and misclassified subjects form meaningful structure in the learned embedding space, and how this compares to raw functional-connectivity representations.





# MOTIVATION

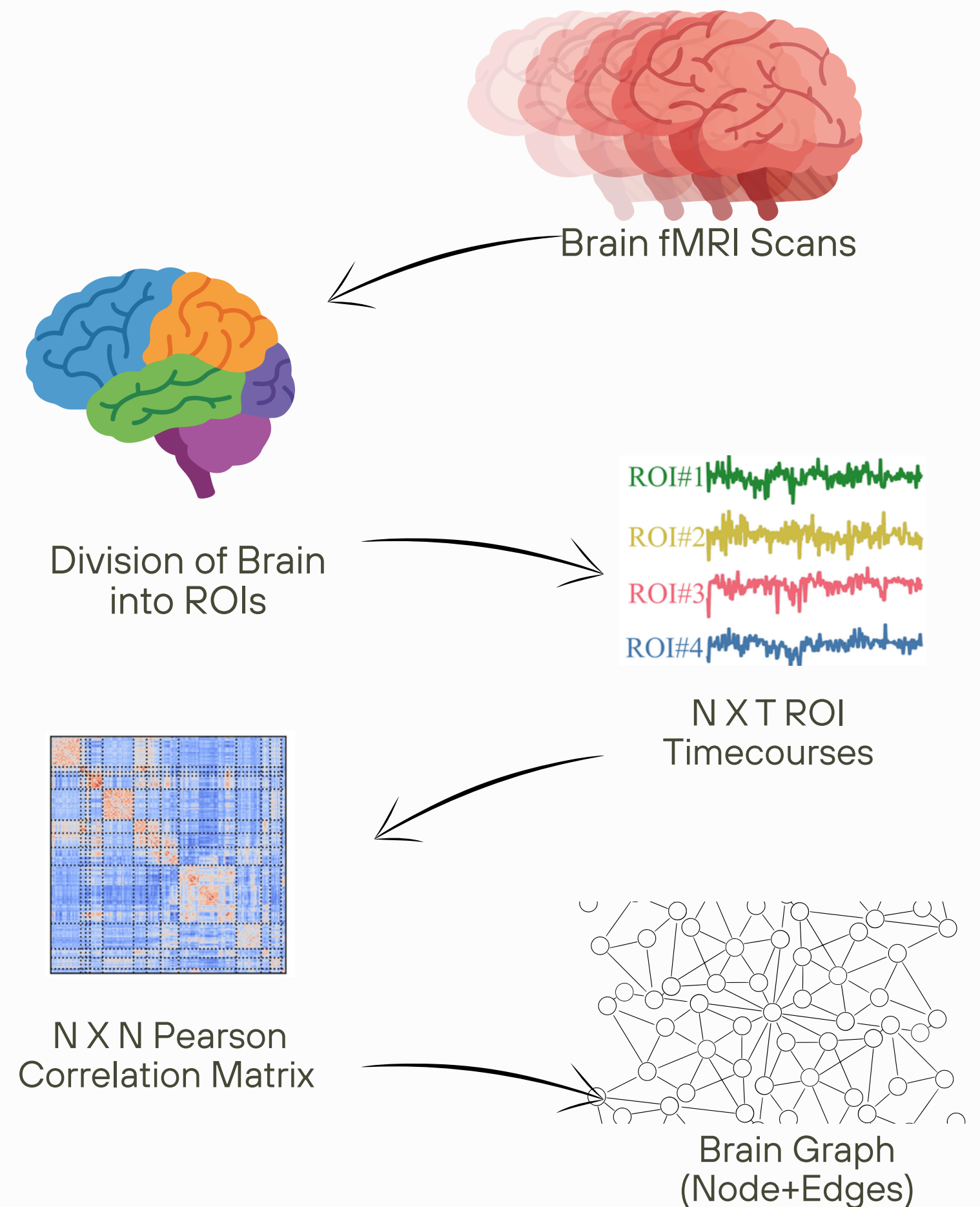
- ASD diagnosis today relies primarily on behavioral observation — it is subjective, often delayed until age 3-4, and dependent on clinician experience
- There is a strong clinical need for an objective, biology-based marker that can support or accelerate diagnosis
- The brain is not a collection of independent regions — it is a network, where cognitive function emerges from coordinated activity across regions
- Functional MRI (fMRI) lets us measure this network activity non-invasively, and studies show ASD is associated with atypical patterns of functional connectivity
- Graph Neural Networks are the natural computational tool here — unlike CNNs or standard deep nets, GNNs can operate directly on network-structured data without forcing it into a grid or vector



Brain Network Analysis

# BRAIN AS A GRAPH

- 4D fMRI scan : Captures brain activity over time as a sequence of 3D volumes (BOLD signal)
- Atlas parcellation : The brain is divided into a fixed set of Regions of Interest (ROIs) using a predefined atlas (this project used AAL, Craddock, and Dosenbach atlases)
- Time series extraction : For each ROI, the average BOLD signal across time is extracted, giving one time series per region
- Pearson correlation : Pairwise correlation between every pair of ROI time series produces a correlation matrix : This becomes the basis of the graph
- Graph construction:
  - Nodes = ROIs (brain regions)
  - Edges = functional connectivity strength (Pearson correlation) between region pairs
  - Correlation matrices are dense and include both positive and negative values, so some form of thresholding or edge-weighting is typically applied
  - This graph is then fed into a GNN for downstream tasks (classification, embedding learning)



# PROBLEM STATEMENT

Autism Spectrum Disorder is associated with a typical patterns of functional connectivity in the brain, which can be captured using resting-state fMRI. A common approach in this area is to represent the brain as a graph, using regions of interest as nodes and Pearson correlation between their activity as edges, and train a Graph Neural Network to classify subjects as ASD or Healthy Control based on this graph. However, several aspects of this pipeline remain open to investigation: whether the fixed, correlation-based graph construction limits model performance, whether GNNs meaningfully leverage network structure compared to simpler baselines, whether a learned, task-aware graph performs better than a fixed one, and what kind of patient representation a trained GNN actually learns. This internship's tasks were structured around these questions, starting with a standard classification setup and progressively moving toward graph construction methods and analysis of the model's learned representations.

## Key Challenges :

- No ground-truth brain graph : The true connectivity structure of the brain is unknown; every graph used is an estimate, and the method used to construct it can affect downstream results
- Small, multi-site clinical data : ABIDE combines scans from multiple sites and scanners with limited subject counts, which increases the risk of overfitting and of site-related variation being mistaken for diagnostic signal
- Unclear benefit of graph structure : Prior work (El Gazzar et al.) shows GNNs do not consistently outperform simpler, structure-agnostic baselines, leaving open the question of how much the graph structure itself contributes





# DATASET - ABIDE

- ABIDE (Autism Brain Imaging Data Exchange) — a multi-site, publicly available rs-fMRI dataset aggregating scans from ASD and Healthy Control subjects across many international sites
- Brain graphs were constructed using three different atlases, each producing a different parcellation (different number/definition of ROIs):
  - AAL (Automated Anatomical Labeling)
  - Craddock
  - Dosenbach
- Each atlas-derived dataset required separate preprocessing: [X.npz] (graph/connectivity data), [Y.npy] (labels), and a demographic\_data.csv (site, age, sex, diagnosis) per atlas
- Subject counts differ across atlases — this is because different atlases led to different amounts of NaN-value filtering during preprocessing (some ROI time series produced invalid correlations depending on parcellation, which had to be dropped)

| Atlas Name | ROI | Total Subject | ASD | HC  | Dropped (NaN filtering |
|------------|-----|---------------|-----|-----|------------------------|
| AAL        | 116 | 825           | 385 | 440 | 8                      |
| Craddock   | 200 | 856           | 398 | 458 | 0                      |
| Dosenbach  | 161 | 825           | 385 | 440 | 0                      |

# LITERATURE REVIEW – I

## Paper 1: Benchmarking GNNs for fMRI Analysis (El Gazzar et al., 2022).

- Systematically evaluated five popular GNN architectures on brain-disorder classification (depression and autism) across multi-site clinical datasets
- Key finding: GNNs did not consistently outperform simpler, structure-agnostic deep learning baselines — a 1D-CNN performed best overall across scenarios
- Identified graph construction itself as the major bottleneck, since existing methods use somewhat arbitrary rules to define edges, resulting in noisy graphs
- Proposed integrating graph diffusion into existing architectures to improve performance
- Relevance to this internship: this paper directly motivated questioning whether the fixed Pearson-correlation graph is actually the right foundation, and shaped the decision to explore learnable graph construction (FBNetGen) later in the internship

## Paper 2: FBNetGen (Kan et al., MIDL 2022).

- Argues that traditional functional brain networks (built via fixed correlation) are noisy and not optimized for any particular downstream prediction task
- Proposes an end-to-end, task-aware framework that learns the brain network structure directly from raw ROI time-series data, rather than using a fixed correlation matrix
- Framework has three components: ROI feature extraction, a learnable graph generator, and a GNN-based predictor — all trained jointly
- Reported improved performance and interpretability (e.g., highlighting task-relevant brain regions) compared to fixed-graph approaches
- This is the paper you implemented yourself — covered in depth on the next slide

## Paper 3: BrainADNet (Barman et al., Communications Medicine 2026).

- Addresses limited training data availability in psychiatric neuroimaging (applied to Major Depressive Disorder) through a brain-signal augmentation strategy
- Built on a Skip-Graph Convolutional Network to aggregate multi-layer features and enrich the model's representational capacity
- Incorporates demographic attributes — age, education, gender — directly into training, recognizing their clinical relevance to diagnosis
- Relevance: introduced the idea that demographic metadata can be a useful signal alongside connectivity data, relevant to the site/sex/age columns tracked in the later latent-space analysis





# LITERATURE REVIEW – II

## Paper 4: GLNSTGNN (Cui et al., iScience 2026)

- Introduces a Hierarchical Sparse Spatiotemporal GNN for brain graph classification, combining graph convolutions with sparsity constraints across both spatial (ROI) and temporal (time-series) dimensions
- Reported accuracy on brain graph classification benchmarks higher than several comparison methods in the paper
- You tested this reported performance context against your own results by applying/evaluating it on the ABIDE dataset, giving a direct point of comparison for your pipeline
- Relevance: represents the more architecturally complex end of the spectrum — useful contrast to the simpler 3-layer GCN used in your own latent-space experiments

## Paper 5: Brain Network Transformer (Kan et al., NeurIPS 2022)

- Moves away from fixed adjacency matrices entirely — models brain networks using a Transformer, using each ROI's full connection profile as its node feature
- Learns pairwise connection strengths between ROIs via attention weights, rather than assuming Pearson correlation defines the graph
- Introduces an Orthonormal Clustering Readout operation that groups ROIs into functional modules in a self-supervised way, producing more interpretable, cluster-aware embeddings
- Standardized its evaluation specifically on the public ABIDE dataset — same dataset used throughout this internship, making it directly comparable
- Relevance: reinforces the theme (alongside FBNetGen) that fixed-graph GCNs may be an outdated default, and attention/learned structure is an active research direction

## Paper 6: Signed Graph Convolutional Network (Derr et al., ICDM 2018)

- Standard GCNs are designed for graphs with only positive edges; this paper proposes a GCN variant that explicitly handles signed networks — graphs containing both positive and negative edges
- Uses balance theory from social network analysis to correctly propagate information across positive and negative links through layers
- Relevance: Pearson correlation matrices in brain graphs naturally contain both positive and negative (anticorrelated) values, which standard GCNs typically discard or treat incorrectly by taking absolute values or thresholding negatives away — this paper's approach directly informed the positive-vs-negative FC investigation carried out later in the internship



# IMPLEMENTING FBNETGEN ON ABIDE

## Motivation:

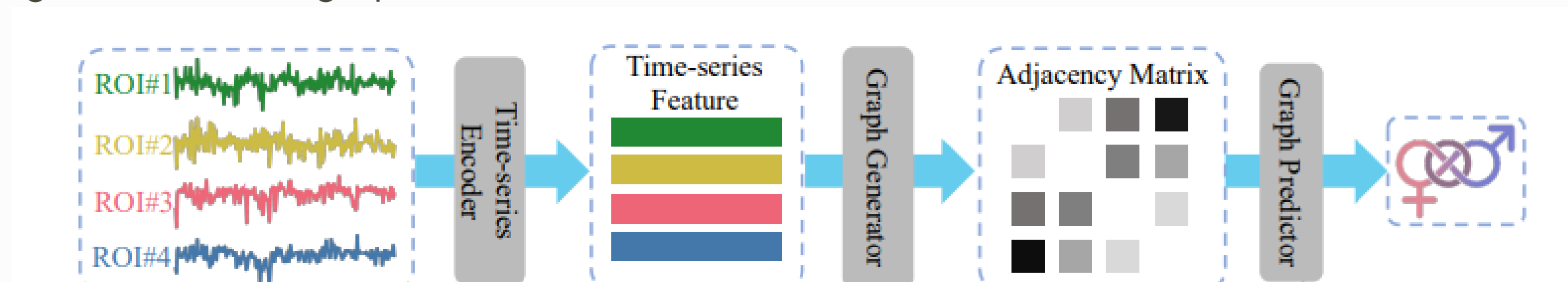
Standard GCNs operate on a fixed, pre-computed adjacency matrix — typically built using static Pearson correlation between ROI time series. This construction step is task-agnostic: it is computed the same way regardless of what the model is ultimately trying to predict, and may therefore fail to capture connectivity patterns that are specifically informative for a given classification objective. This limitation, highlighted in the benchmarking literature reviewed earlier, motivated exploring an alternative graph construction strategy.

## Approach:

FBNetGen was implemented as an end-to-end, task-aware alternative to fixed-graph construction. Instead of starting from a pre-computed correlation matrix, the framework learns the brain network structure directly from raw fMRI time-series data. The graph generation step is optimized jointly with the downstream classification objective, meaning the resulting adjacency matrix is shaped by what is actually predictive of ASD versus HC, rather than fixed in advance by a statistical rule.

## Implementation details:

- An existing GCN codebase was reused as the framework's final predictor module — a standard and time-efficient engineering choice, since this stage functions as a conventional GCN once the graph has been generated
- The components upstream of the predictor were built specifically for this implementation: a time-series encoder to extract features from raw ROI signals, a learnable graph generator module (implemented via `main_fbnetgen.py` and `utils_fbnetgen.py`) that outputs the adjacency matrix, and an integration layer connecting the generated graph into the reused GCN predictor
- The complete pipeline was applied to the ABIDE dataset



# LATENT SPACE STUDY — SETUP

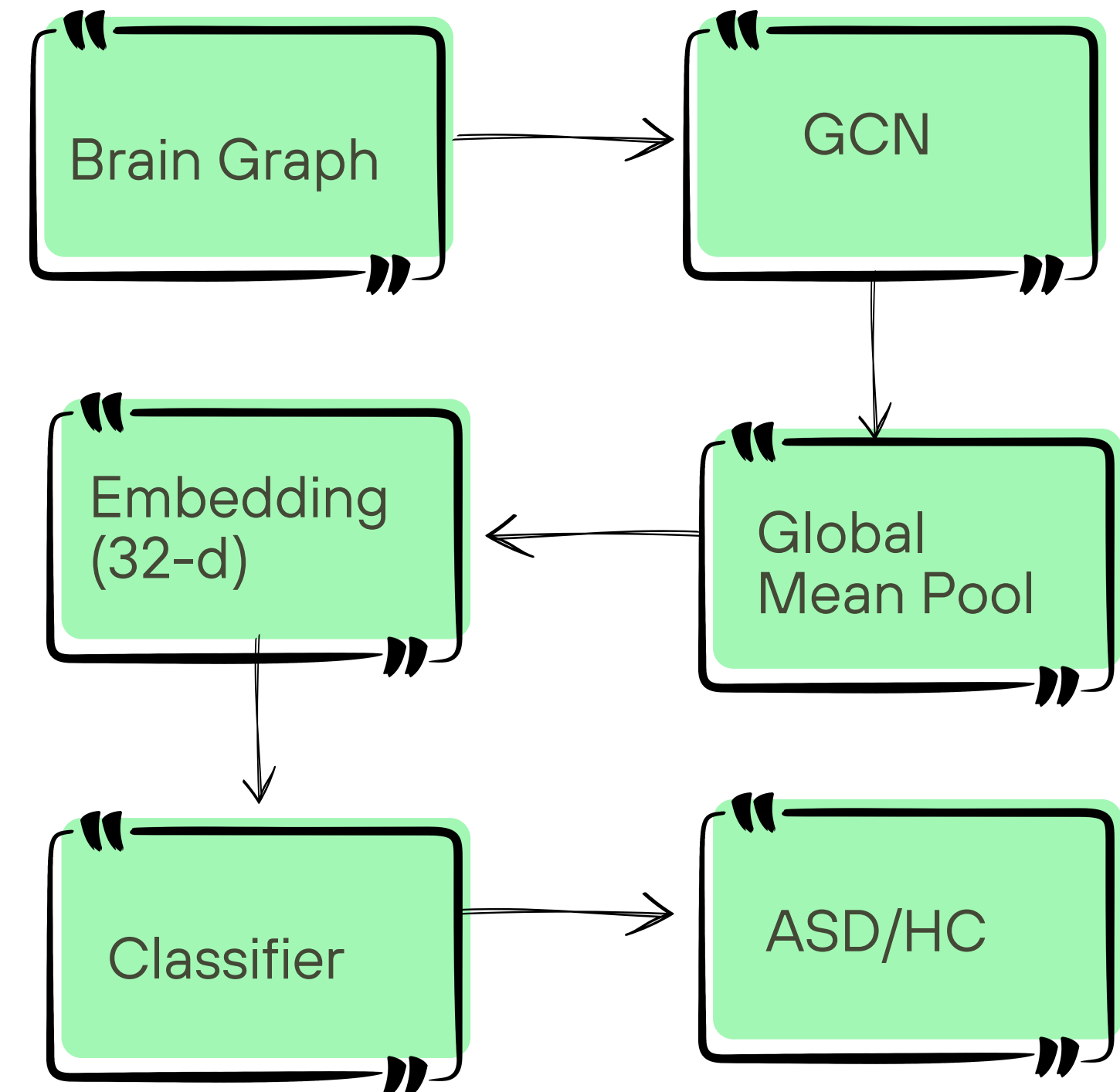
Rather than stopping at classification accuracy, this task examines the internal representation a trained GCN builds for each subject — the 32-dimensional embedding produced at global pooling, just before the final classification layer. A 3-layer GCN ( $116 \rightarrow 128 \rightarrow 64 \rightarrow 32$ ) was trained on AAL-atlas brain graphs, using Fold 1 of a 10-fold stratified split (661 train / 74 val / 82 test), with the lowest-validation-loss checkpoint used for all analysis that follows.

## Metrics

Accuracy  
F1 Score  
ROC-AUC  
Sensitivity  
Specificity

## Value

0.622  
0.652  
0.656  
0.659  
0.579

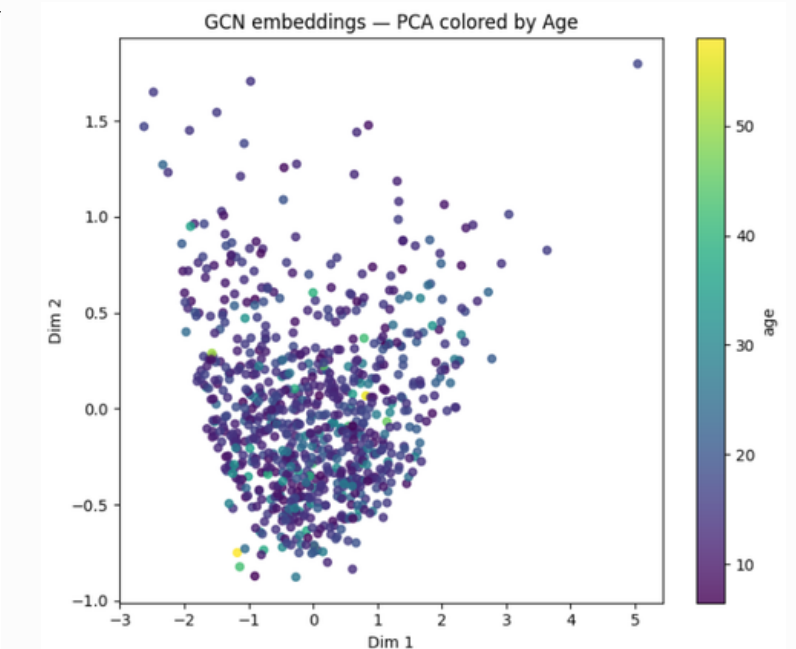
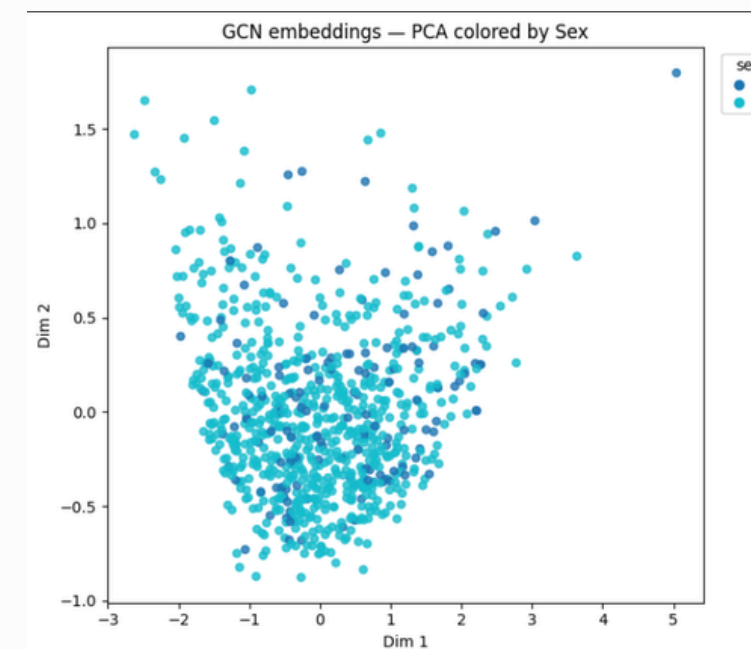
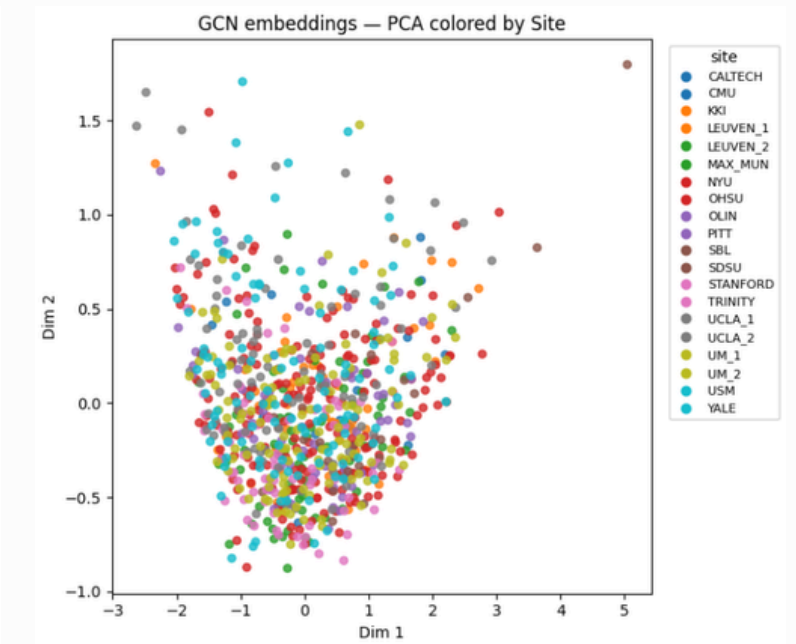
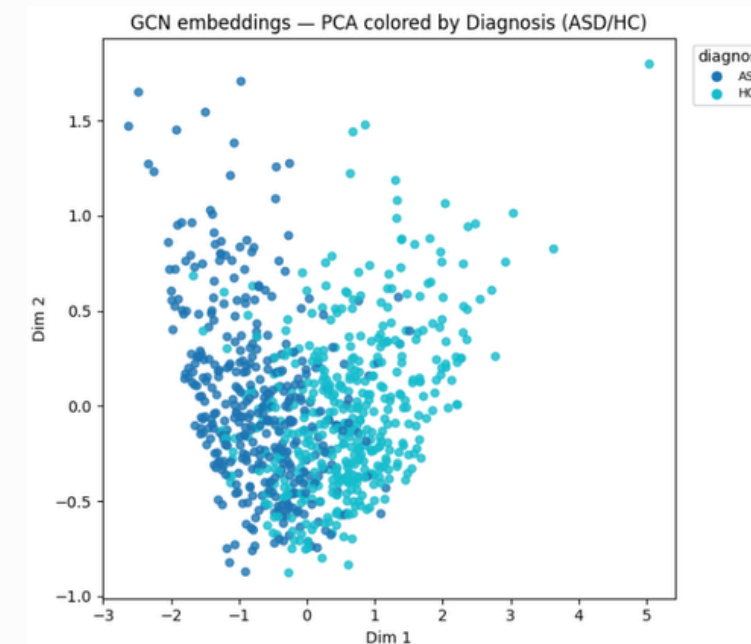




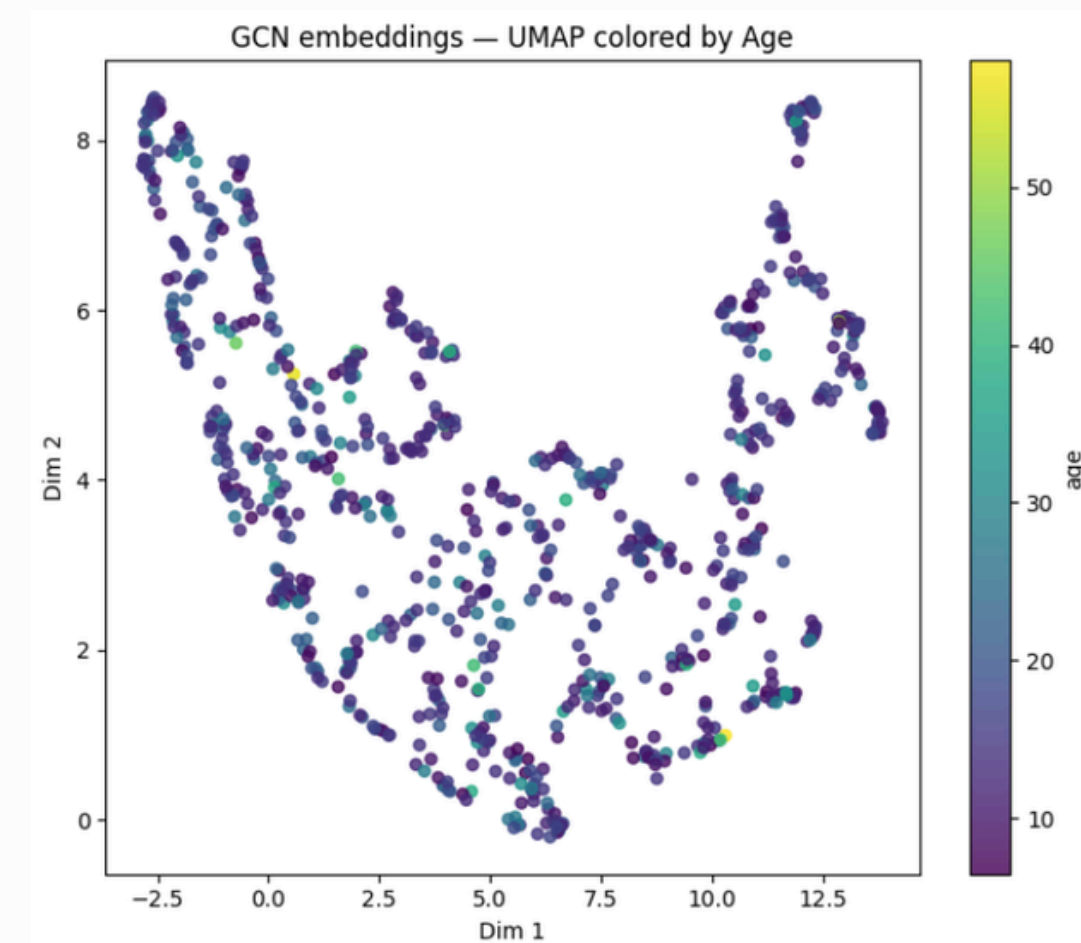
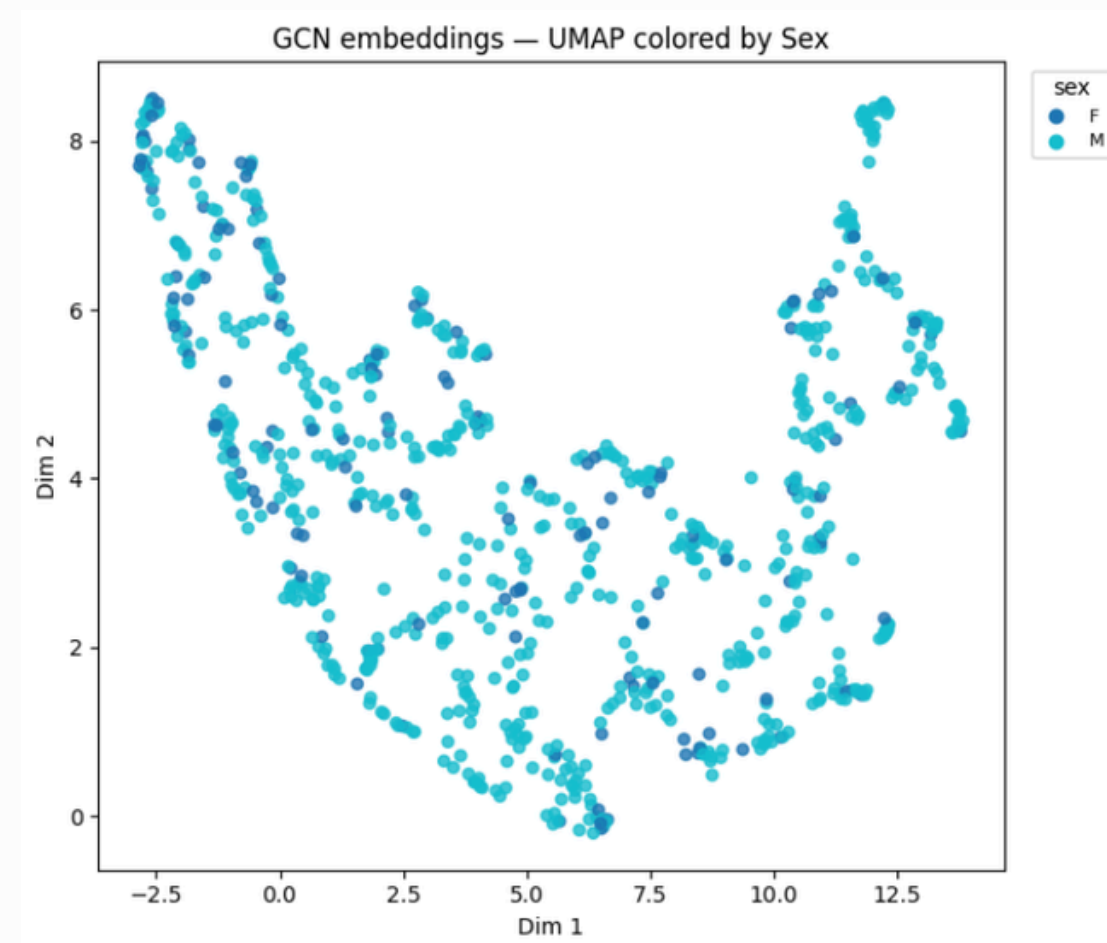
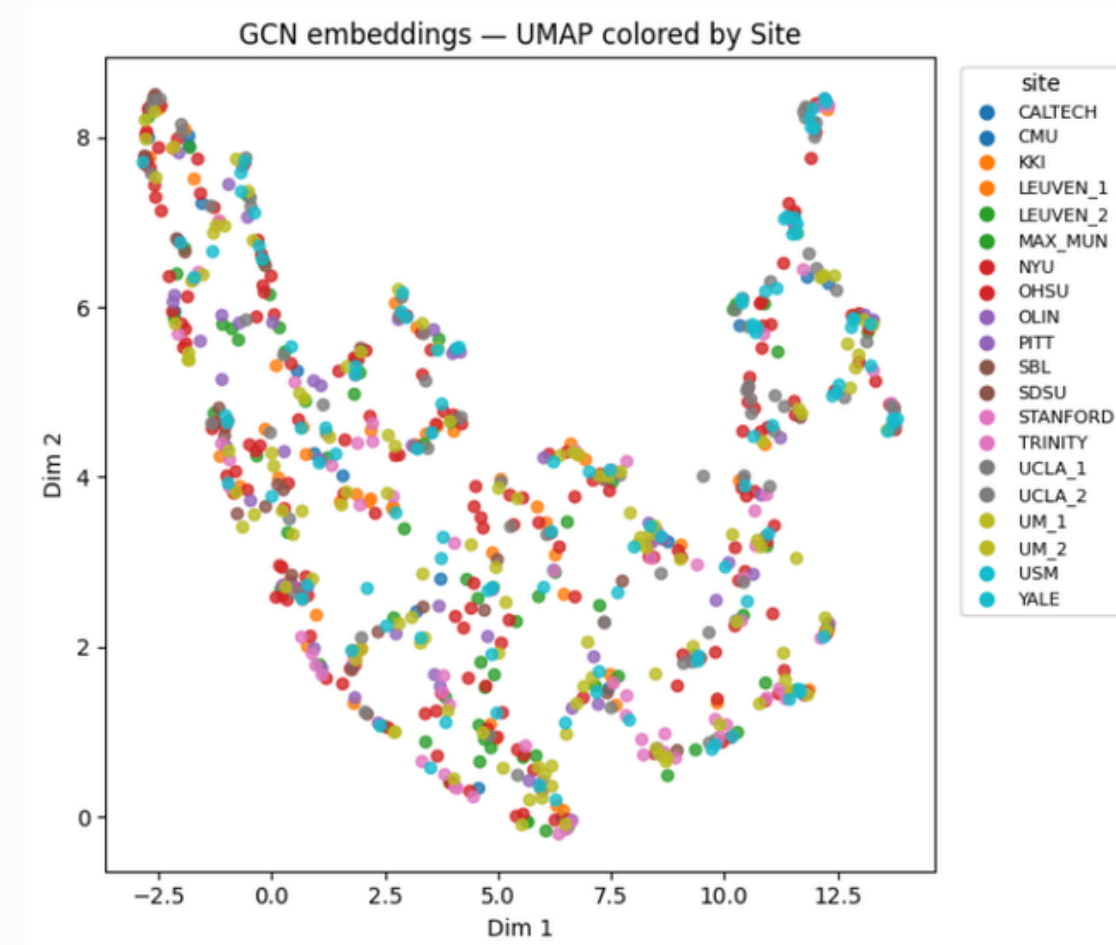
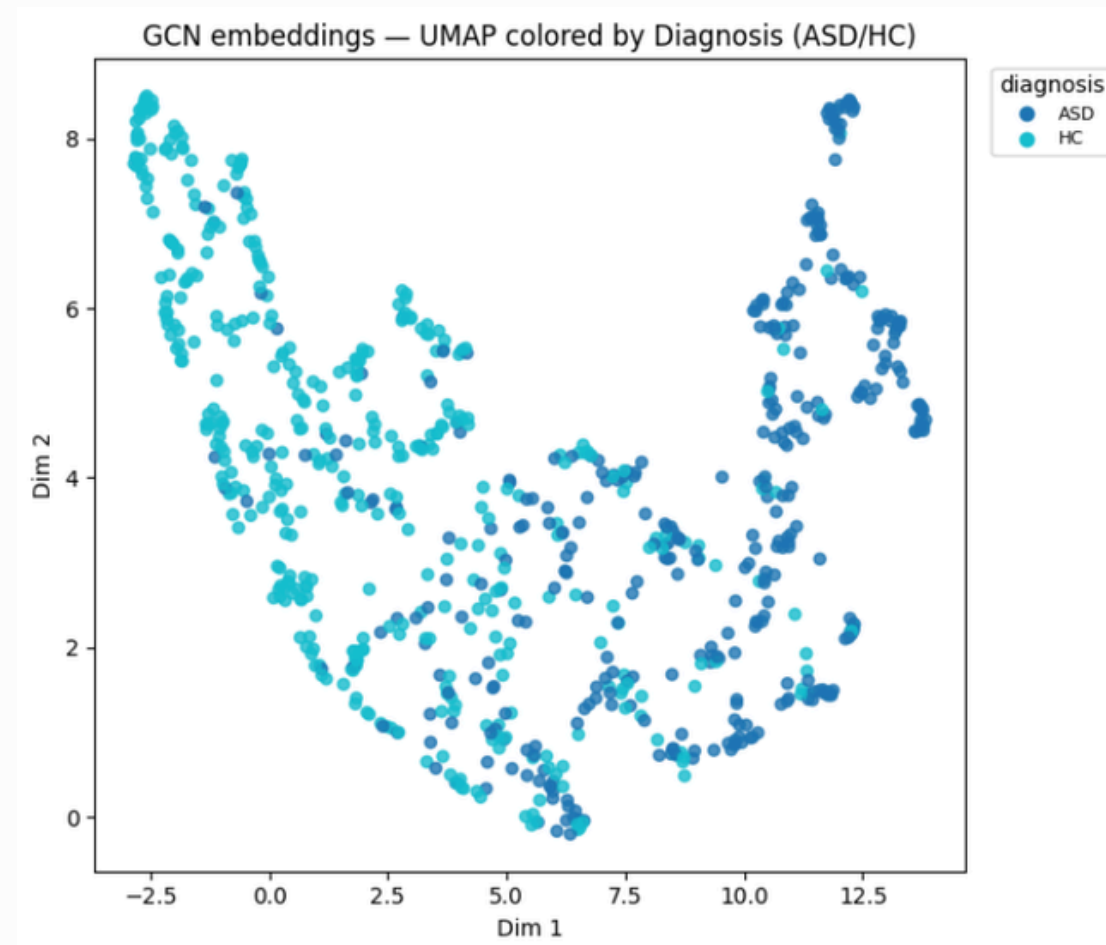
# VISUALIZING THE LEARNED EMBEDDING SPACE

PCA's first two components explain  $\sim 99.5\%$  of variance, meaning the embedding is dominated by very few directions rather than spread across all 32 dimensions.

The genuine concern: ASD and HC subjects overlap heavily in both PCA and UMAP — the model has not learned a representation where the two classes separate. This is the direct explanation for why classification accuracy stayed around 62%: a model can't be much more accurate than its embedding space allows. Site shows partial, branch-wise clustering under UMAP — a mild but real site effect worth flagging, since it means part of what the model captures may be scanner identity rather than diagnosis. Sex and age showed no meaningful structure.

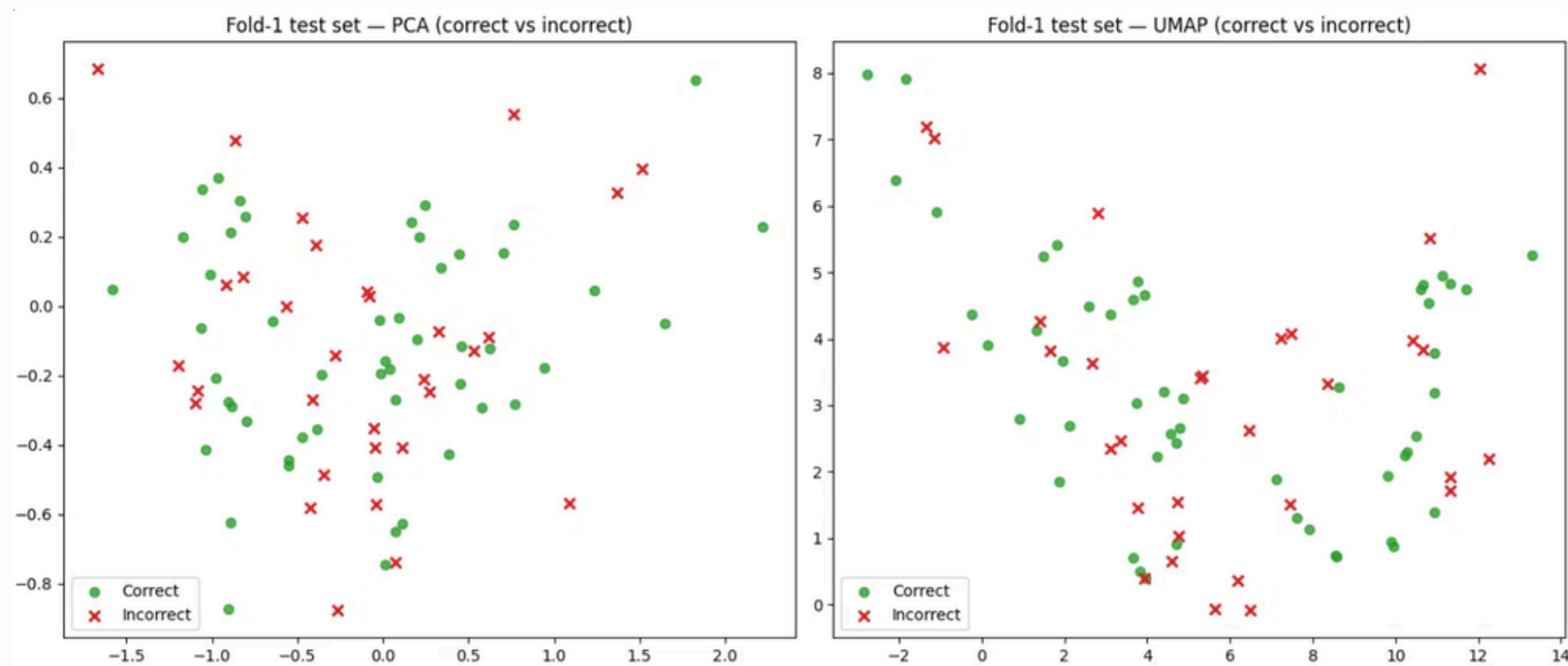




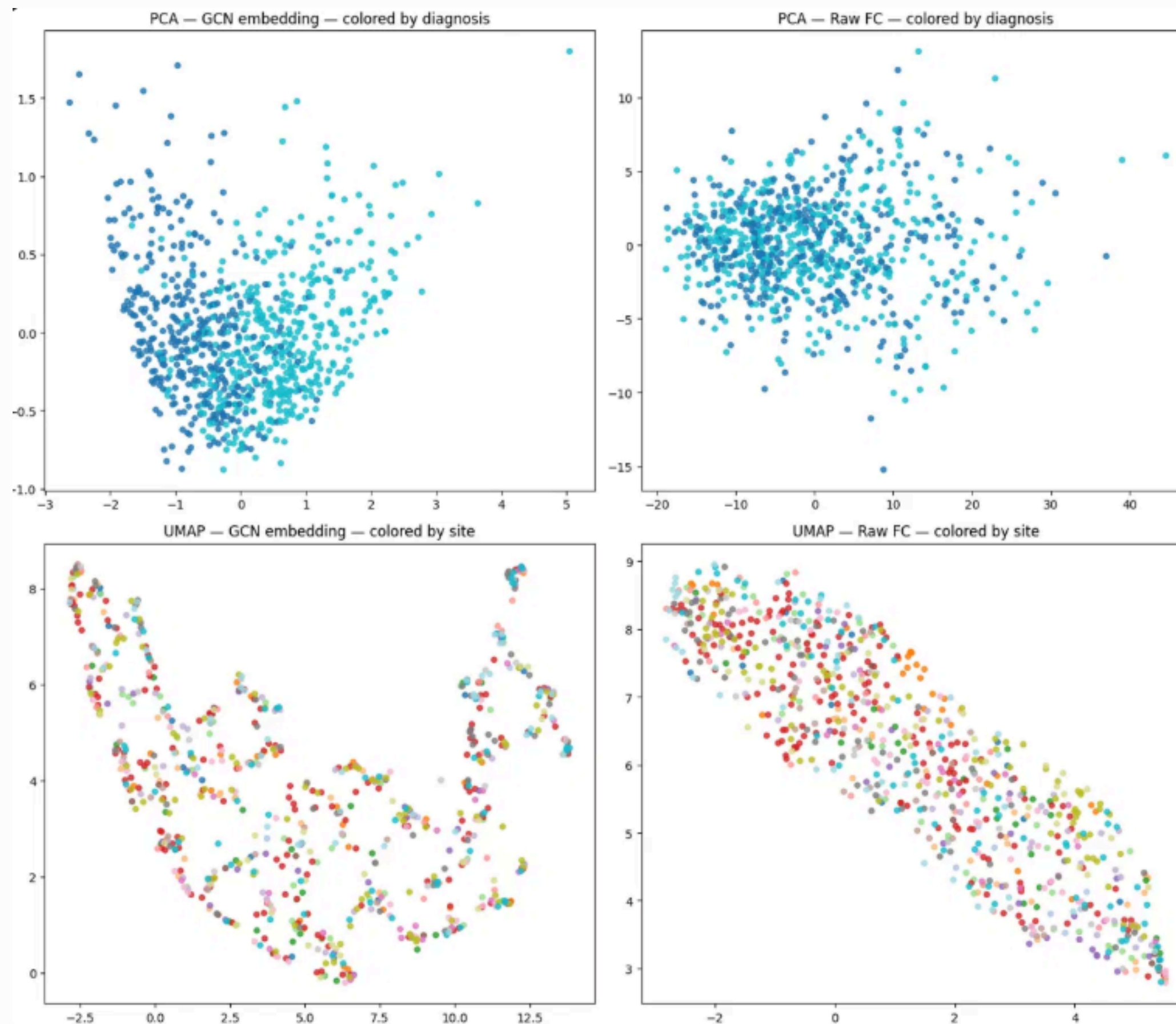


# ERROR ANALYSIS & COMPARISON WITH RAW CONNECTIVITY

Misclassified test subjects are scattered across the space, not concentrated at an ASD/HC boundary — expected, since the two classes weren't cleanly separated to begin with. No site was disproportionately represented among the errors.



The real test is whether the GCN adds anything over the raw data. Raw Pearson-correlation vectors (no GCN) were projected the same way. Neither the GCN embedding nor raw FC showed a clearer ASD/HC split — meaning the network, at this scale, did not learn a more diagnostically useful representation than the input already contained. This is an important, honest result: it points to graph construction or model capacity — not the presence of a GNN — as the likely bottleneck, echoing the exact concern raised in the benchmarking paper reviewed earlier.





# REFERENCES & GITHUB

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Full code, results, and task reports available at:  
<https://github.com/ArihaGupta15/IIT-Delhi-Internship-2026>



**THANK YOU  
VERY MUCH**